

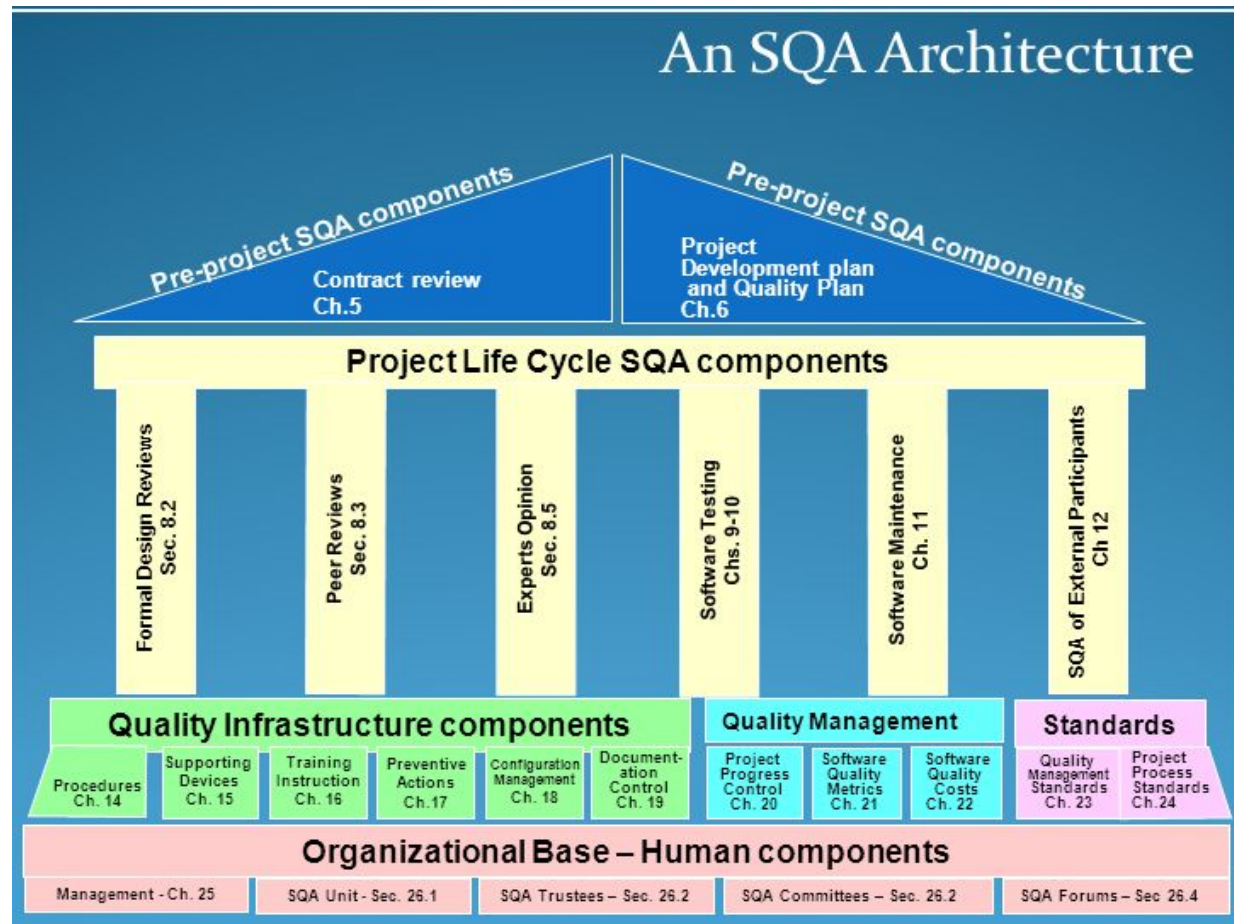


Unit 5

Contract Review

Recap- The SQA System

- An SQA system always combines **a wide range of SQA components.**



Recap- The SQA System

- All are employed to challenge the multitude of sources of software errors and to achieve an acceptable level of software quality.
- **SQA system components can be classified into 6 classes :**
 - **Pre-project** quality components
 - **Project life cycle** quality components
 - **Infrastructure** (organisation wide) error prevention and improvement quality components
 - **Software** quality management components
 - **Standardisation, certification and SQA** assessment components
 - **The human SQA** components

Pre-project Quality Components

- Pre-project quality components relate to:
 1. Contract reviews, and
 2. Project development plans and quality plans



Contract Review

- Software development **may** be executed within the framework of a contract negotiated with a customer or in response to an internal order originating in another department.
- SQA efforts in this space aim at preventing the creation of **bad contracts** that may yield low-quality software.
 - *usually characterised by loosely defined requirements, and unrealistic budgets and schedules*

Contract Review

- The contract review process originates **in the customer–supplier relationship**, but is expected to make a substantial contribution to internal projects as well.
- In both instances, the development unit is committed to an *agreed-upon functional specification, budget and schedule*.
 - *What, how much, how long*



Contract Review

- This unit is dedicated to the study of **the objectives of contract review** and **the wide range of review subjects** that correspond to these objectives.
- Contract review activities must include a detailed examination of:
 - (a) the **project proposal draft** and
 - (b) the **contract drafts**.



Contract Review

Unit Case Example: CFV Project

○ **Good News**

- Successful completion of a 10-month project for Carnegie Fruits and Vegetables, a produce wholesaler.
- The project was conducted in full as originally scheduled.
- Each member had received a nice bonus for finishing on time.

Unit Case Example: CFV Project

○ **Bad News**

- The very successful project had actually lost about \$90 000.
- The estimates failed to account for client training requirements
- “Nobody tried to find out how many branches the client operated. Nobody mentioned that CFV operated 19 branches – six of them overseas – before signing the contract!”
- The sales negotiating team was blamed for the loss.

Unit Case Example: CFV Project

- **Recipe for project or service failure?**
 - Sloppily written proposals
 - Poorly understood contracts
 - Shallow and quick resource estimates
 - Exaggerated software sales efforts
 - Unrealistic schedules and budgets
 - Unrealistic professional commitments



THE CONTRACT REVIEW PROCESS AND ITS STAGES

Why Sign a Contract?

- Several situations can lead a SW company to sign a contract with a customer such as:
 - Participation in a **tender**
 - Submission of a proposal according to the customer's **RFP**.
 - *Generally a tender request is for finding a vendor for services at the lowest price. Whereas a RFP is for a vendor with the best service regardless of price.*
 - Receipt of an order from a company's customer
 - Receipt of an internal request or order from another department in the organisation.

The Contract review process and its stages

○ **Contract review :**

- is the SQA component devised to guide the *review* of **proposal** and **contract** document drafts.
- If applicable, provides oversight (supervision) of the contracts carried out with potential project partners and subcontractors.

Contract Review Stages

- **Stage 1** – Review of the proposal draft prior to submission to the potential customer (proposal draft review):
- Reviews the final proposal draft and proposal's *foundations*:
 - Customer's requirement documents
 - Customer's additional details and explanations of the requirements
 - Cost and resources estimates
 - Existing contracts or contract drafts of the supplier with partners and subcontractors.



Contract Review Stages

- **Stage 2** – Review of the contract draft prior to signing (Contract draft review):
 - Reviews the contract draft on the basis of the proposal and the understandings (including changes) reached during the contract negotiation sessions.



The Contract review process

- The individuals who perform the review **thoroughly examines the draft while referring to a comprehensive range of review subjects.**
 - **A Check-list** is very helpful for assuring the full coverage of relevant subjects (see appendix 5A, 5B).

Appendix 5A Excerpt: **Objective** - Customer requirements have been clarified and documented

Subjects

1. The functional requirements.
2. The customer's operating environment (hardware, data communication system, operating system, etc.).
3. The required interfaces with other software packages and instrument firmware, etc.
4. The performance requirements, including workloads as defined by the number of users and the characteristics of use.
5. The system's reliability.

Check-list 5A

- 9. Definition and protection of software proprietary rights
 - 9.1 Securing proprietary rights to software purchased from others.
 - 9.2 Securing proprietary rights to data files purchased from others.
 - 9.3 Securing proprietary rights to future reuse of software developed in custom-made projects.
 - 9.4 Securing proprietary rights to software (including data files) developed by the firm (the supplier) and his subcontractors during the development period and while in regular use by the client.

Appendix 5B Excerpt: **Objective** - No unclarified issues remain in the contract draft

○ **Subjects**

- Supplier's obligations as defined in the contract draft and its appendices.
- Customer's obligations as defined in the contract draft and its appendices.



The Contract review process

- After the completion of a review stage it is required that:
 - the necessary changes, additions and corrections are introduced by the proposal team (after the proposal draft review) and
 - by the legal department (after the contract draft review).

Contract Review Activities

- *Clarification* of the customer's requirements
- *Review* of the project's schedule and resource requirement estimates
- *Evaluation* of the professional staff's capacity to carry out the proposed project
- *Evaluation* of the customer's capacity to fulfil his obligations
- *Evaluation* of development risks



PROPOSAL & CONTRACT REVIEW OBJECTIVES

Proposal Draft Review objectives

- The objectives assure the following:
 - [1] Customer requirements have been clarified and documented**
 - In the event that RFP documents are too general and imprecise, additional details should be obtained from the customer.
 - Clarifications should be recorded and approved by both the customer and the software firm.
 - [2] Alternative approaches for carrying out the project have been examined**
 - Such as software reuse, and partnerships or subcontracting with firms that have specialised knowledge or staff that can qualify for meeting the proposal's terms.

Proposal Draft Review objectives

[3] Formal aspects of the relationship between the customer and SW firm have been specified.

- To include:
 - Customer communication and interface channels
 - Project deliverables and acceptance criteria
 - Formal phase approval process
 - Customer design and test follow-up method
 - Customer change request procedure.



Proposal Draft Review objectives

[4] Identification of development risks

- such as insufficient professional know-how regarding the project's professional area or the use of required development tools, need to be identified and resolved

[5] Adequate estimation of project resources and timetable have been prepared.

- To include professional staff as well as the project's budget, including subcontractors' fees.



Proposal Draft Review objectives

- [6] Examination of the firm's capacity with respect to the project.**
 - Relating to professional competence as well as the availability of the required team members and development facilities on the scheduled time.

- [7] Examination of the customer's capacity to fulfill his commitments**
 - refers to the customer's financial and organisational capacities, such as personnel recruitment and training, installation of the required hardware, and upgrading of its communications equipment.



Proposal Draft Review objectives

[8] Definition of partner and subcontractor participation conditions

- Covers quality assurance issues, payment schedules, distribution of project income/profits, and cooperation between project management and teams.

[9] Proprietary rights have been defined and protected

- In cases where reused software is inserted into a new package or when rights for future reuse of the current software need to be decided.

Contract Review Objectives

Contract draft review objectives (assure the following)

[1] No un-clarified issues remain in the contract draft

- See appendix 5B

[2] All the understandings reached between the customer and the firm are to be fully and correctly documented.

[3] No changes, additions, or omissions that have not been discussed and agreed upon should be introduced into contract draft.



IMPLEMENTATION OF A CONTRACT REVIEW



What Factors determine the extent of a contract review?

- Contract reviews vary in their magnitude, depending on the characteristics of the proposed project.
- This complexity may be either technical or organisational.

What Factors determine the extent of a contract review?

[1] Project magnitude

- usually measured in Man(person)- -month resources
- Man-month represents the productive effort of one person in a 4-week period.

[2] Project technical complexity

- This includes all technical aspects of the project, such as;
 - Number of technologies involved
 - Familiarity of team with technologies
 - Cutting edge or well established technology
 - Number of technical interfaces

What Factors determine the extent of a contract review?

[3] Degree of staff acquaintance with and experience in the project area.

- In cases where a high proportion of software reuse is possible, the extent of the review is reduced.

[4] Project organisational complexity,

- the greater the number of organisations (partners, subcontractors, and customers) taking part in the project, the greater the contract review efforts required.



Who performs a contract review?

- The leader or another member of the proposal team
- The members of the proposal team
- An outside professional or a company staff member who is not a member of the proposal team.
- A team of outside experts.

Implementation of a contract review of a major proposal

- The characteristics of the major proposal:
 - Very large-scale project
 - Very high technical complexity
 - New professional area for the company
 - High organisational complexity
- The difficulties of carrying out contract reviews for major proposals:
 - Time pressures
 - Proper contract review requires substantial professional work
 - The potential contract review team members are very busy

Implementation of a contract review of a major proposal

• Recommended avenues (approaches) for implementing major contract reviews :

- The contract review should be scheduled.
- A team should carry out the contract review
- A contract team leader should be appointed
- The activities of the team leader include :
 - Recruitment of the team members
 - Distribution of review tasks
 - Coordination between members
 - Coordination between the review team and the proposal team
 - Follow-up of activities, especially compliance with the schedule
 - Summarisation of the findings and their delivery to the proposal team.

Contract review for internal projects

Type of internal project	The in-house customers	Project examples
(1) Administrative or operative software to be applied internally	Administration and operating units	■ Sales and inventory systems ■ Financial resource management systems ■ Human resource management systems
(2) Software packages originally intended to be sold to the public as “off-the-shelf” packages	Software marketing department	■ Computer games ■ Educational software ■ Word processors ■ Sales and inventory management software packages
(3) Firmware to be embedded in the company’s products	Electronic and mechanical product development departments	■ Electronic instrumentation and control products ■ Household amusement equipment and machinery ■ Advanced toys

Contract review for internal projects

- The main point here is the internal relationship –
 - Best described as loose and based on general agreements, with goodwill playing an important role between the two units
 - Hence, internal projects tend to be characterised by insufficient examination of the project's requirements, its resources and development risks.

Disadvantages of “loose relationships” internal projects

Subject	Disadvantages to the internal customer	Disadvantages to the internal developer
(1) Inadequate definition of project requirements	■ Implementation deviates from needed applications ■ Low satisfaction	■ Higher than average change requirements ■ Wasted resources due to introducing avoidable changes
(2) Poor estimate of required resources	■ Unrealistic expectations about project feasibility	■ Substantial deviations from development budget ■ Friction between units induced by requirements for budget additions
(3) Poor timetable	■ Missing scheduled dates for beginning distribution of new products	■ Development activities are under time pressures and tend to suffer from low quality ■ Late project completion causes delays in freeing staff for their next project
(4) Inadequate awareness of development risks	■ Customer unprepared for project risks and their consequences	■ Tardy initiation of efforts to overcome difficulties



Contract review for internal projects

- To avoid the problems of internal projects, actions include defining:
 - An adequate proposal for the internal project
 - Applying a proper contract review process
 - An adequate agreement between the internal customer and the internal supplier.